



B-NINE®

WATER SOLUBLE GRANULE PLANT GROWTH REGULATOR FOR ORNAMENTALS

Active Ingredient: (% by wt.)

daminozide [butanedioic acid mono
(2,2-dimethylhydrazide)]..... 85.0%

Other Ingredients: 15.0%

Total: 100.0%

EPA Reg. No. 70506-532-101563

KEEP OUT OF REACH OF CHILDREN CAUTION / PRECAUCION

If you do not understand the label, find someone to explain it to you in detail. (Si usted no entiende la etiqueta, busque a alguien para que se la explique a usted en detalle.)

**See inside panel for First Aid Instructions
and Booklet for complete Precautionary
Statements and Directions for Use.**

envu

MANUFACTURED FOR
Environmental Science U.S., LLC
5000 CentreGreen Way, Suite 400
Cary, NC 27513

FIRST AID

IF IN EYES	• Hold eye open and rinse slowly and gently with water for 15 to 20 minutes. • Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eye. • Call a poison control center or doctor for treatment advice.
IF SWALLOWED	• Call a poison control center or doctor immediately for treatment advice. • Have person sip a glass of water if able to swallow. • Do not induce vomiting unless told to do so by a poison control center or doctor. • Do not give anything by mouth to an unconscious person.
IF INHALED	• Move person to fresh air. • If person is not breathing, call 911 or an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. • Call a poison control center or doctor for further treatment advice.
IF ON SKIN OR ON CLOTHING	• Take off contaminated clothing. • Rinse skin immediately with plenty of water for 15 to 20 minutes. • Call a poison control center or doctor for treatment advice.

HOTLINE NUMBER

Have the product container or label with you when calling a poison control center or doctor, or going for treatment.

FOR EMERGENCY MEDICAL ASSISTANCE, CALL 1-800-424-9300

FOR CHEMICAL EMERGENCY: Spill, leak, fire, exposure, or accident call CHEMTREC 1-800-424-9300.

FOR PRODUCT USE INFORMATION: CALL 1-800-331-2867

PRECAUTIONARY STATEMENTS

HAZARDS TO HUMANS AND DOMESTIC ANIMALS

CAUTION: Causes moderate eye irritation. Avoid contact with eyes or clothing. Wash thoroughly with soap and water after handling.

PERSONAL PROTECTIVE EQUIPMENT

Applicators and Other Handlers Must Wear: Long-sleeved shirt and long pants; waterproof gloves; shoes plus socks.

Follow manufacturer's instructions for cleaning and maintaining PPE. If no such instructions for washables, use detergent and hot water. Keep and wash PPE separately from other laundry. When handlers use closed systems or enclosed cabs in a manner that meets the requirements listed in the Worker Protection Standard (WPS) for agricultural pesticides [40 CFR 170.607(d-e)], the handler PPE requirements may be reduced or modified as specified in the WPS.

USER SAFETY RECOMMENDATIONS

Users should:

- Wash hands before eating, drinking, chewing gum, using tobacco, or using the toilet.
- Remove clothing immediately if pesticide gets inside. Then wash thoroughly and put on clean clothing.

(continued)

PRECAUTIONARY STATEMENTS (continued)

ENVIRONMENTAL HAZARDS

Do not contaminate water when disposing of equipment washwaters.

PHYSICAL OR CHEMICAL HAZARDS

Do not use or store near heat or open flame.

DIRECTIONS FOR USE

It is a violation of Federal law to use this product in a manner inconsistent with its labeling. Read all label directions carefully before use.

Do not apply this product through any type of irrigation system.

Do not apply this product in a way that will contact workers or other persons, either directly or through drift. Only protected handlers may be in the area during application. For any requirements specific to your State or Tribe, consult the agency responsible for pesticide regulation.

AGRICULTURAL USE REQUIREMENTS

Use this product only in accordance with its labeling and with the Worker Protection Standard, 40 CFR Part 170. This Standard contains requirements for the protection of agricultural workers on farms, forests, nurseries, and greenhouses, and handlers of agricultural pesticides. It contains requirements for training, decontamination, notification, and emergency assistance. It also contains specific instructions and exceptions pertaining to the statements on this label about personal protective equipment (PPE), and restricted-entry interval. The requirements in this box only apply to uses of this product that are covered by the Worker Protection Standard.

Do not enter or allow worker entry into treated areas during the restricted-entry interval (REI) of 12 hours.

PPE required for early entry to treated areas that is permitted under the Worker Protection Standard and that involves contact with anything that has been treated, such as plants, soil, or water, is:

- Coveralls over long-sleeved shirt and long pants;
- Chemical-resistant gloves – made of any waterproof material; such as Barrier Laminate, Butyl Rubber ≥ 14 mils, Nitrile Rubber ≥ 14 mils, Neoprene Rubber ≥ 14 mils, Natural Rubber ≥ 14 mils, Polyethylene, Polyvinyl Chloride (PVC) ≥ 14 mils or Viton ≥ 14 mils.
- Shoes plus socks.

PRODUCT INFORMATION

B-NINE® is a plant growth regulator for use on container grown ornamental plants. When applied to the plant foliage **B-NINE** passes into the leaf, then moves within the plant to the growing points to reduce internode elongation. Treated plants will develop a shorter, more compact growth habit when compared to untreated plants.

FOR USE ONLY ON ORNAMENTALS GROWN IN COMMERCIAL OR RESEARCH GREENHOUSES, SHADEHOUSES AND NURSERIES. IN PRODUCTION AREAS NOT UNDER COVER, USE IS RESTRICTED TO CONTAINERIZED ORNAMENTALS¹.

MIXING INSTRUCTIONS

Use the DILUTION TABLE below to determine the amount of **B-NINE** needed for the required concentration:

CONCENTRATION OF B-NINE DESIRED		VOLUME MEASURE* FOR ONE GALLON OF SOLUTION		WEIGHT MEASURE FOR ONE GALLON OF SOLUTION	
%	PPM	Level Scoops	Level Tablespoons	Ounces (avoirdupois)	Grams
0.100	1,000	4/5	4/5	0.16	4.5
0.125	1,250	1	1	0.20	5.6
0.250	2,500	2	2	0.40	11.1
0.375	3,750	3	3	0.60	16.7
0.500	5,000	4	4	0.80	22.3
0.750	7,500	6	6	1.20	33.4

*Approximate dosage only. Use weight measurements wherever possible.

Place the required amount of **B-NINE** into the spray tank. Then add the amount of water needed to make the desired concentration. Allow 3 to 5 minutes for **B-NINE** to dissolve in water, stirring occasionally. Once the **B-NINE** has dissolved in water, no additional agitation is required.

Solutions prepared from **B-NINE** should not be held for more than 24 hours.

Be sure to seal the unused portion of **B-NINE** in its original container as high humidity and moisture will cause it to harden. Do not tank mix **B-NINE** with copper-containing products and do not apply **B-NINE** within 7 days either before or after application of such products, as burning of leaves may result. Do not add additional wetting agent or mix with insecticides, fungicides or foliar fertilizers, as burning of leaves may result.

APPLICATION NOTES

For best results apply 1 gallon of **B-NINE** spray solution to cover 200 square feet of area. This will provide foliage coverage just to the point of runoff.

Plants must be well watered prior to application of **B-NINE**. Wilting plants do not readily absorb **B-NINE** and poor results might be obtained if application is made to wilted plants.

The plant foliage must be dry when sprays are applied. Do not syringe or overspray the foliage for 18 to 24 hours after applying **B-NINE** to allow the chemical to enter the plant.

Two applications of **B-NINE** at 1/2 to 2/3 the label rates may give better control than a single application of the total labeled rate.

¹ Use on bed-grown or containerized ornamentals without impervious floors not permitted by the Arizona Department of Environmental Quality.

B-NINE TANK MIX WITH CHLORMEQUAT CHLORIDE: On certain crops where **B-NINE** alone provides limited height control, the application of this tank mix can provide a synergistic effect that results in enhanced activity. Optimum rates for the tank mix can be adjusted to obtain a desirable level of activity, but the general guidelines are to use a chlormequat chloride rate which does not cause phytotoxicity and to then vary the rate of **B-NINE** to adjust activity.

It is the pesticide user's responsibility to ensure that all products are registered for the intended use. Read and follow the applicable restrictions and limitations and directions for use on all product labels involved in tank mixing. Users must follow the most restrictive directions for use and precautionary statements of each product in the tank mixture.

The following table provides general guidelines for rates of **B-NINE** based on the desired level of activity. Use the labeled rate of the chlormequat chloride product based on phytotoxicity concern.

ACTIVITY LEVEL	B-NINE (ppm)
Very High	5,000
High	2,500
Medium	1,250
Low	800

USE AND RATE DIRECTIONS BY CROP

Although the use of **B-NINE** has been found to be acceptable for the specific plants listed in this label, be sure to read and fully understand the section on DETERMINING PLANT SAFETY before applying to large numbers of plants.

A. AZALEAS (FLORIST)

To promote compactness and development of additional buds on plants grown for forcing, spray when new growth following final shaping is 2 to 3 inches with a 0.25% (2500 ppm) solution of **B-NINE**.

An additional application may be made two weeks later, if necessary.

B. BEDDING PLANTS / FLOWERING PLANTS (Annual or Perennial)

Excessive stem elongation of most bedding plants is prevented by a **B-NINE** spray made when 1 1/2 to 2 inches of new growth has developed after transplanting or pinching (approximately 2 to 3 weeks) or when the plants begin to elongate (stretch). Use of a 0.25% to 0.50% (2500 to 5000 ppm) concentration is particularly effective on species such as, but not limited to:

Ageratum	Crossandra	Kalanchoe
Aster	Dahlia	Lisianthus
Begonia	Delphinium	Marigold
Browallia	Dianthus	Petunia
Calendula	Dusty Miller	Phlox
Celosia	Exacum	Salvia
Centaurea	Gerbera	Verbena
Coleus	Gomphrena	Vinca
Cosmos	Hibiscus	Zinnia

Repeat applications may be made when new growth begins to stretch.

To control the length of the peduncle (flower stalk) on plants such as Kalanchoe and Gerbera, apply **B-NINE** at a concentration of 0.12% to 0.50% (1200 to 5000 ppm). For Gloxinia, do not exceed 0.125% (1250 ppm) or injury may occur. Begin applications 2 to 3 weeks after initiation of short days.

Note: Some varieties of Kalanchoe with cupped leaves have shown a sensitivity to **B-NINE** that results in slight phytotoxicity to the foliage. This phytotoxicity occurs when **B-NINE** spray solution accumulates, and therefore concentrates, in these cupped areas. To avoid possible phytotoxicity on such varieties, apply **B-NINE** solution during the late afternoon or evening; rinse the foliage thoroughly with water the next morning.

To reduce leaf injury of petunias caused by air pollutant ozone and sulfur dioxide: Make a second application of 0.50% (5000 ppm) one week after the first. **B-NINE** will not protect against leaf damage caused by the air pollutant PAN. Growth retardation will be more pronounced and there may be a delay in flowering time from the second **B-NINE** application. Plants sprayed for pollution protection will be no shorter than those sprayed for compact growth.

C. BEDDING PLANT PLUGS

In order to maintain proper size of bedding plant plugs for transplanting and shipping, apply **B-NINE** to plugs as needed, beginning at the first true leaf stage (III) through the finishing stage (IV).

For optimum growth control, apply **B-NINE** at concentrations of 0.15% to 0.25% (1500 to 2500 ppm). The concentrations used will depend on seedling vigor and projected number of applications. Repeat applications can be made at 1 to 2-week intervals or as needed. The most vigorous seedlings may require weekly applications.

Properly treated plugs will not be overly stunted and will continue to grow when transplanted.

In addition to the bedding plant species listed above (Section B), **B-NINE** has also been shown to be effective in controlling the height of impatiens and pansies in the plug stage.

D. CHRYSANTHEMUMS (POT)

To produce compact, branched (pinched) potted plants, apply **B-NINE** in the range of 0.25% to 0.5% (2500 to 5000 ppm). For very sensitive varieties, apply the 0.25% rate when the desired height is reached. For other cultivars, use the 0.25% rate during winter and 0.5% in the summer, when axillary shoots are 1 to 2 inches long.

Additional applications, if required, may be made at 2-week intervals, stopping one week prior to disbudding.

It is suggested that a "MEDIUM" schedule calling for shading and pinching on the same day be used for "TALL" varieties. This will eliminate the open, relatively leafless appearance often associated with the use of delayed pinch schedules.

E. CHRYSANTHEMUMS (CUT)

To retard neck stretching just below the flower on potted and cut flowers, spray the top 1/3 of the foliage to runoff with 0.25% (2500 ppm) **B-NINE** two days after disbudding.

F. CHRYSANTHEMUMS (CUTTINGS)

At certain times of the year or with certain tall or long internoded varieties, the conventional spray applications described above may not be adequate. Additional applications of **B-NINE** may be useful as pre-plant dips and post-plant sprays as follows:

1. Pre-plant dip: - Use 0.10% (1000 ppm):

ROOTED CUTTING: Hold a handful of cuttings by the root system; dip in the **B-NINE** solution just long enough to thoroughly wet all leaves and stems, remove and pot. Allow the foliage to dry before watering-in. Normally, by the time you are at the end of a bed, you can start watering-in those plants first potted.

UNROOTED CUTTING: Remove unrooted cuttings from the box. Dip stem in the **B-NINE** solution, remove and place in flat. Cover with cheesecloth to avoid dehydration and place in the cooler overnight. Stick the next day and go on to a normal misting program.

2. Post-plant spray: Spray rooted cuttings as soon as the cuttings establish themselves (approximately 3 to 5 days after potting).

Concentration will vary from 0.125% to 0.50% (1250 to 5000 ppm), depending on variety and time of year.

G. FOLIAGE PLANTS

B-NINE, sprayed at a concentration of 0.25% to 0.75% (2500 to 7500 ppm) is effective for controlling elongation (stretching) and promoting improved appearance and coloration on foliage plants such as, but not limited to:

Ficus	Pothos	Schefflera
Philodendron	Radermachera	Syngeonim

Two or more applications may be necessary if new growth begins to stretch or for enhanced coloration.

H. GARDENIAS

To produce compact plants for spring bloom, spray with a 0.50% (5000 ppm) **B-NINE** when plants are about 2/3 of final marketing size. This is from mid-August to mid-September for the Florida area and during the first two weeks of December for most other areas.

I. HYDRANGEAS

To retard spring forcing growth, spray with a 0.50% (5000 ppm) solution of **B-NINE** when the new growth has just started to unfold and 4 to 5 pairs of leaves are visible. Apply 2 to 4 weeks after the start of forcing but not later than this time.

For those greenhouse varieties that produce excessive growth during the summer following pinching, spray to runoff with a 0.75% (7500 ppm) solution of **B-NINE** when the regrowths are 1 to 2 inches in length, usually in late July, but not after the first week in August.

J. POINSETTIAS

To reduce the elongation (stretching) of poinsettias, begin application of **B-NINE** when new growth is 1 1/2 to 2 inches long.

Seasonably late applications will reduce plant height but may also reduce bract size and delay flowering. For growers scheduling early December flowering, applications should not be made after initiation of short days. As a guide, do not apply after October 1 for areas outside Florida, or after October 25 in Florida.

Use a rate range of 0.20% to 0.30% (2000 to 3000 ppm). Use the 0.2% dilution in areas north of the sunbelt and 0.3% in the sunbelt. Vigorous growing varieties may require higher rates and repeat applications may be necessary for optimum plant size.

DETERMINING PLANT SAFETY

The use of **B-NINE** has been found to be acceptable for the specific plants listed on this label. Due to the large number of species and varieties of ornamentals, it is impossible to test every one for efficacy and safety to **B-NINE**. Neither the manufacturer nor the seller has determined whether or not **B-NINE** can be used safely on ornamentals not specified on this label. Prior to any large scale application on such plants, the user must determine the efficacy and safety of **B-NINE** by testing a small number of the type of plants to be treated at the labeled rates for that particular group, i.e. bedding plants, foliage, etc., and observing the treated plants over a one to two week period for phytotoxicity. The user assumes all risks arising out of application to unlabeled plants.

STORAGE AND DISPOSAL

Do not contaminate water, food or feed by storage or disposal.

PESTICIDE STORAGE: Store in the original container only. After partial use, close the container tightly. Store in a secure place that is temperate and dry. High humidity and moisture will cause this product to harden. Use spray and stock solutions within 24 hours. Do not store solutions. Immediate use is appropriate if another component is added to spray mixture.

PESTICIDE DISPOSAL: Wastes resulting from the use of this product may be disposed of on site or at an approved waste disposal facility.

CONTAINER HANDLING

Nonrefillable container. Do not reuse or refill this container. Offer for recycling, if available.

Plastic or paper bags: Completely empty bag into application equipment, then offer for recycling if available or dispose of empty bag in a sanitary landfill, or by incineration.

**IMPORTANT INFORMATION
READ BEFORE USING PRODUCT**

**CONDITIONS OF SALE AND LIMITATION
OF WARRANTY AND LIABILITY**

NOTICE: Read the entire Directions for Use and Conditions of Sale and Limitation of Warranty and Liability before buying or using this product. If the terms are not acceptable, return the product at once, unopened, and the purchase price will be refunded.

The Directions for Use of this product reflect the opinion of experts based on field use and tests, and must be followed carefully. It is impossible to eliminate all risks associated with the use of this product. Crop injury, ineffectiveness or other unintended consequences may result because of such factors as manner of use or application, weather or crop conditions, presence of other materials or other influencing factors in the use of the product, which are beyond the control of Environmental Science U.S., LLC or Seller. Handling, storage, and use of the product by Buyer or User are beyond the control of Environmental Science U.S., LLC and Seller. To the extent consistent with applicable law, all such risks shall be assumed by Buyer and User, and Buyer and User agree to hold Environmental Science U.S., LLC and Seller harmless for any claims relating to such factors.

TO THE EXTENT CONSISTENT WITH APPLICABLE LAW, ENVIRONMENTAL SCIENCE U.S., LLC AND SELLER MAKE NO WARRANTIES OF MERCHANTABILITY OR OF FITNESS FOR A PARTICULAR PURPOSE NOR ANY OTHER EXPRESS OR IMPLIED WARRANTY EXCEPT AS STATED ON THIS LABEL.

To the extent consistent with applicable law, Environmental Science U.S., LLC or Seller shall not be liable for any incidental, consequential or special damages resulting from the use or handling of this product and **THE EXCLUSIVE REMEDY OF THE USER OR BUYER, AND THE EXCLUSIVE LIABILITY OF ENVIRONMENTAL SCIENCE U.S., LLC AND SELLER FOR ANY AND ALL CLAIMS, LOSSES, INJURIES OR DAMAGES (INCLUDING CLAIMS BASED ON BREACH OF WARRANTY, CONTRACT, NEGLIGENCE, TORT, STRICT LIABILITY OR OTHERWISE) RESULTING FROM THE USE OR HANDLING OF THIS PRODUCT, SHALL BE THE RETURN OF THE PURCHASE PRICE OF THE PRODUCT OR, AT THE ELECTION OF ENVIRONMENTAL SCIENCE U.S., LLC OR SELLER, THE REPLACEMENT OF THE PRODUCT.**

Environmental Science U.S., LLC and Seller offer this product, and Buyer and User accept it, subject to the foregoing conditions of sale and limitations of warranty and of liability, which may not be modified except by written agreement signed by the duly authorized representative of Environmental Science U.S., LLC.

B-NINE is a trademark of a UPL Corporation Limited Group Company.

Envu and the Envu logo are trademarks owned by Environmental Science U.S., LLC or one of its affiliates.

MANUFACTURED FOR
Environmental Science U.S., LLC
5000 CentreGreen Way, Suite 400
Cary, NC 27513

envu™